A review of SIROCCO 2012

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Who would have thought that Sirocco winds could one day blow over Iceland? The distributed computing community made it happen by holding the 19th International Colloquium on Structural Information and Communication Complexity (SIROCCO) in Reykjavik[1].

With its 120,000 inhabitants, its harbor, small town houses, but crowded bars and many cultural venues, Reykjavik has an air of a village and one of a European metropolis at the same time. In July, the days last forever and so does the hustle and bustle in Reykjavik streets. Reykjavik University, where the conference took place, is located in the southern part of the city, in a newly built campus on the shore, just next to a naturally heated sand beach, and is equipped with modern and pleasant facilities. The fifty or so participants had there the perfect environment to discuss advances in SIROCCO's areas of interest.

SIROCCO is devoted to the study of communication and knowledge in distributed systems from both qualitative and quantitative viewpoints. It has been indeed the aim of SIROCCO, since its 1994 first edition, to comprehend trade-offs and interplay between information that entities are provided in a distributed system and their computational and communication power. Aside from rather traditional problems such as agreement, local decision or graph exploration, new and innovative ones were also given special attention such as problems involving game theoretic aspects and quantum computations. And it is in reflection of this innovative spirit that each year, a Prize for Innovation in Distributed Computing is awarded to individuals whose contribution introduced new approaches or research areas.

This year, 28 talks were selected among 54 submitted papers, with Roger Wattenhofer and Boaz Patt-Shamir as invited speakers.

Boaz Patt-Shamir (Tel Aviv University) gave a lecture on a new type of online set packing (OSP), where large data frames are broken into packets assuming that in each step, only one packet can be served by a router, and all other packets are lost. He presented a randomized distributed algorithm for maximizing the number of useful data frames, for which none of

[1]Previous locations include Greece, Italy, France, Switzerland, Spain, and more recently Slovenia ('09), Turkey ('10) and Poland ('11).

their constituent packets are lost, showing how the basic approach extends to other models. The talk was based on various papers co-authored with Yuval Emek, Magnús Halldórsson, Yishay Mansour, Jaikumar Radhakrishnan, and Dror Rawitz.

Roger Wattenhofer (ETH Zurich) was the recipient of this year's Prize for Innovation in Distributed Computing, in recognition to his contribution to the study of distributed approximation. Wattenhofer initiated and developed the study of distributed approximation on many key problems such as *facility location*, *minimum dominating set* or *graph coloring*. He also investigated related areas including LP-based techniques for approximation and local approximation, thus improving the understanding of locality. In his talk, Wattenhofer addressed the topic of Distributed Complexity Theory, presenting some lower bounds on finding the diameter of a graph, approximating a minimum vertex cover, and many other problems. He also introduced some exciting new research directions, such as the networked Finite State Machine model.

The talks

The session chairs strictly timed the 75 minute sessions, which were interleaved with coffee or lunch breaks featuring some of the most delicious Icelandic food[2].

Here is a brief overview of SIROCCO's scientific contribution. Please do refer to the LNCS proceedings [3] for more complete material.

Graph related problems :

When it comes to distributed systems, graph related problems are *de rigueur*, with their share of connectivity, orienting, routing or coloring questions.

For instance, Y. Emek et al. give new insights into the notion of connectivity in multi-layer networks and shed light on deep differences with the regular single-layer model, where three notions of connectivity are equivalent. They show that one of these notions can indeed be computed in polynomial time in the multi-layer model while the two other notions have been shown to be hard to compute or approximate in this model.

I. Gamzu and M. Medina presented a paper improving the approximation ratio for the maximum mixed graph orientation problem where the objective is to orient the undirected edges of a mixed graph so as to maximize the number of pairs that admit a directed source-target path.

H. Hasemann et al., by addressing a particular scheduling problem - namely the fractional graph coloring problem - show that there exist classical graph problems that cannot be solved locally if the local outputs are required to be of a constant size. Moreover, these problems require that each node has a unique numerical identifier. This result points out a loophole in the usual definitions of scheduling problems that can be exploited in the design of local distributed algorithms.

[2]The author of this review strongly recommends the Hjónabandssæla (traditional Icelandic cookie, with oatmeal and jam) to any sweet craving person.

[3]Volume 7355, 2012, DOI: 10.1007/978-3-642-31104-8

Addressing the routing problem, I. Caragiannis and C. Kalaitzis consider the greedy paradigm that uses an embedding of the nodes into points of a metric spaces equipped with distance function. They show that greedy embeddings may have high stretch (ratio between routing path and the optimal one), and that optimal stretch requires the use of a $\Omega(n)$-dimension space. This result disproves a conjecture by Maymounkov stating that greedy embeddings of optimal stretch are possible with polylogarithmic dimension.

Modeling wireless dynamic networks is challenging (see below for more about wireless networks), thus making it tricky to transpose traditional graph results to this model. Concerning the *agreement* problem for example, M. Biely, P. Robinson and U. Schmid use dynamic directed graphs to model this type of networks and prove that consensus is impossible under some weak connectivity assumptions. They nonetheless introduce vertex-stable root component concept to circumvent this impossibility.

Wireless networks :

One of the hot topics of this conference was the Signal-to-Interference-and-Noise-Ratio model. The SINR model attempts to recreate the *physical* model of wireless communications, where interference and noise are taken into account to describe the loss of signal between a sender and a receiver. E.I. Ásgeirsson, M.M. Halldórsson and P. Mitra look into the stability of such wireless networks: they give a distributed algorithm that achieves $\Omega(\frac{1}{\log^2 n})$-efficiency in maintaining the queue sizes bounded when processors are faced with stochastic arrivals of packets. As for D. Yu et al., they study the multiple-message broadcasting problem in the SINR model and give an $O(D+k+\log^2 n)$ randomized protocol to deliver any message m over the network, where D is the network diameter and k the number of overlapping messages. They also prove that any uniform randomized algorithm needs $\Omega(D+k+\frac{\log^2 n}{\log\log\log n})$ timeslots to deliver k messages initially stored at k nodes to all the nodes in the network.

Still regarding the interferences in wireless networks, M. Khabbazian, S. Durocher and A. Haghnegahdar show how to solve the interference minimization problem efficiently in settings typical for ad hoc wireless networks. The interference minimization problem is to assign a transmission radius to nodes such that the resulting communication graph is connected, while minimizing the maximum interference.

Two talks addressed the wireless sensor networks model. The first one, presented by Oscar Morales-Ponce, studies sensor networks with double antennae and gives algorithms for orienting the antennae so as to attain *strong connectivity* using optimal angular range. As for A. Bar-Noy, B. Baumer and D. Rawitz, they present a polynomial-time algorithm for 2-duty cycling over the strip cover problem, where sensors are grouped into shifts of at most 2 agents, and must take turns covering a one-dimensional region.

Fault tolerance and Self-stabilization :

To be resilient in an error prone environment is vital for distributed systems. It is hence essential to design algorithms that cope with such errors.

C. Delporte, H. Fauconnier and H. Tran-The address the problem of Byzantine Agreement when faulty processors may forge identities creating a *homonymy* situation in the system. They show that if at most t processors may be Byzantine and at most k among the set of l identifiers are forgeable, then the problem is solvable if and only if $l > 2t+k$. They also extend

this result to the case where the processors are authenticated by signatures. Another work related to the identities has been presented, by D. Alistarh et al.: regarding the renaming problem, namely when n processors need to choose identifiers from a namespace of limited size, the authors show that by adapting the running time to the number of failures f in the execution, the problem can be solved in constant time if f is limited to $O(\sqrt{n})$, and that in general the problem can always be solved in $O(\log f)$ rounds. They also give a tight lower bound of $\Omega(\log n)$ for the wait-free case (when $f = n - 1$).

X. Vilaça, O. Denysyuk and L. Rodrigues consider the problem of reliability transferring data from a set of producers to a set of consumers under the BAR model (where processors may be Byzantine, Altruistic, or Rational). They present an algorithm for asynchronous systems. Their problem also addresses the problem of collusion, be it induced by Byzantine or Rational agents.

S. Chechik and D. Peleg present two fault tolerant versions of the well known capacitated k-center problem, where k service centers are to be located so as to minimize the maximum distance from a client vertex and its assigned service center. If a failure occurs in some of the centers, all nodes (resp. only affected nodes for the second version) are allowed to be reassigned to alternate centers. The paper presents a 9-approximation (resp. 17-approximation) polynomial-time algorithm to solve the problem under this new assumption.

On the stability and failure recovery subject, A. Larsson and P. Tsigas present a self stabilizing (k, r)-clustering algorithms within wireless ad-hoc networks (k is is the number of cluster heads and r is the maximum number of communication for any node to reach a cluster head). J. Chalopin, Y. Mtivier and T. Morsellino look into the stable properties of a distributed system that can be computed anonymously with local snapshots.

Finally, M. Raynal and J. Stainer enrich two distributed asynchronous computation models (Base read/wright model and Iterated immediate snapshot model) with failure detectors classes, and present a simulation algorithm that proves that the two models are computationally equivalent for wait-free task solvability with these failure detectors.

Mobile computing :

On the subject of graph exploration and random walks, K. Altisen et al. looked into random walks with tabu lists: the authors define a class of random walks using lists that contain already visited vertices (tabu lists). They give a characterization of the update rules of these lists ensuring the performance of the walk. S. Dobrev, R. Královič and E. Markou investigate the trade-off between the amount of advice available to a mobile agent and the quality of an online graph exploration, giving an $\Omega(n \log n)$ bound on the necessary number of bits of advice to achieve a competitive ratio 1 with respect to an optimal algorithm knowing the topology of the network. Another interesting result is given by B. Balamohan et al. in their paper about asynchronous graph exploration where they give tight bounds on the number of pebbles and on the size of the team of mobile agents necessary to locate a black hole in a graph.

Another key problem in the mobile agents setting is the *rendezvous* problem. E. Elouasbi and A. Pelc as well as S. Kawai et al. explore some aspects of randomization, regarding respectively the time of anonymous rendezvous in trees and randomized rendezvous in anonymous unidirectional rings. As for the multiple agents case, G. D'Angelo et al. study the gathering

problem on grid networks and give full characterization about gatherable configurations. A paper entitled Getting Close without Touching by L. Pagli, G. Prencipe and G. Viglietta rises the attendant problem of *near-gathering*, presenting a collision-free algorithm for this problem.

... and others :

As said before, one of SIROCCO characteristics is its variety of subjects. No wonder that trying to categorize all the talks in separate closed areas is vain. Here are some other exciting work presented during the conference that doesn't fall in any of the previous categories:

- P. Kling, F. Meyer auf der Heide and P. Pietryzk consider the *online facility leasing* problem where clients arrive over time and are to be assigned to some facility center. Opening a facility causes a certain cost and assures the facility to be functional for a certain period of time. The authors give the first online algorithm for this problem that has time-independent competitive factor.
- V. Bilò, M. Flammini and V. Gallotti couple game theory with graphs - viewed as social knowledge graphs - by studying bidimensional congestion games. On the same game theoretic note, M. Flammini et al. introduce a new class of network creation games, called mobile network creation games, modeling the spontaneous creation of communication networks by mobile devices.
- H. Arfaoui and P. Fraigniaud address the question "what can be computed without communication?" in the case of two agents, and explore the power of quantum entanglement as a resource. They show that apart from games using the XOR function as a final decision function, quantum correlations do not help to solve two-player games with better probability than what can be achieved with classical resources.

Beyond the talks

Despite this long list of interesting talks, a conference wouldn't be so enjoyable without the traditional social events that allow the researchers to come together in a more casual environment.

On the evening of the first day, the participants gathered in the third floor of Mars (the University buildings are named after the solar system) to attend a stand-up dinner. It was an excellent opportunity for everybody to get to discuss in a relaxed atmosphere. Some even ventured to have a sing-along, accompanied by Ýmir Vigfússon on the piano.

On the afternoon of the second day, a bus took us on a fascinating tour through the Reykjanes peninsula. We first stopped by the Álfagjá rift valley, where we crossed the bridge between the American and the Eurasian tectonic plates. Then we had the chance to observe some species of sea-birds and to see Iceland's oldest lighthouse at Reykjanesviti, followed by a quick visit to the Krisuvik geothermal area. Later on, we soaked ourselves in the geothermal spa of the Blue Lagoon, probably the most famous attraction of the region.

But the high point of this excursion was without any doubt the private dinner party we had that evening at Magnús Halldórsson's place. Sports fans watched the football European championship final opposing Italy to Spain (our Italian fellows were not very pleased with the outcome), while the



Figure 1: The Blue Lagoon.

rest of us enjoyed the dinner in a warm cosy atmosphere. Once again, entertainment was provided around the piano. Distributed computing scientists are quite the singers.



Figure 2: The SIROCCO crowd on a bird cliff and later on watching the game.

Was it the location, the flawless organization or the scientific content? The 2012 edition of SIROCCO was a hit. I, for one, am looking forward to the 20th edition that will take place in Ischia, Italy. Let's bet it won't be any less exciting.

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All photo credits go to Stefan Dobrev, entitled official photographer of SIROCCO'12 .

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